Log-Distance Integrability for Dissipative Flows: BV Selections on Fractal Attractors

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Abstract

We introduce a regularity framework for nearest-point projections onto compact attractors of dissipative semiflows in Hilbert spaces. The classical Lipschitz projection theorem (Federer, 1959) requires positive reach—a condition incompatible with fractal attractor geometry. We replace it with a *trajectory log-Lipschitz* condition (TLL), which controls the projection modulus at the geometrically natural scale of the attractor distance rather than the partition mesh. Combined with a *log-distance integrability* condition (LDI), this yields bounded-variation selections along absolutely continuous trajectories—precisely the regularity used by Gronwall-type arguments in conditional regularity criteria. The framework is formulated for dissipative semiflows with a compact attractor of finite fractal dimension; for the 3D Navier–Stokes equations it should be read as a conditional interface for a strong or preselected trajectory-attractor framework. Reaction-diffusion systems and damped wave equations provide cleaner test classes. Two verifiable sufficient conditions for LDI are identified: bounded variation of the log-distance function, and power-law sublevel-time control.

1 Introduction

Let H be a separable Hilbert space and $\mathcal{A} \subset H$ a compact set—the global attractor of a dissipative semiflow $\{S(t)\}_{t\geq 0}$. A fundamental question in the qualitative theory of dissipative PDEs is the regularity of the nearest-point projection

$$P(x) \;\in\; \arg\min_{a\in\mathcal{A}} \|x-a\|_H$$

as a function of time along a trajectory $u(t) = S(t)\,u_0$. If the map $t \mapsto P(u(t))$ is absolutely continuous (AC), Gronwall-type energy estimates can be closed under the companion hypotheses; if it is merely measurable, the same error terms are not controlled by this route.

The classical theory (Federer [10]) guarantees Lipschitz continuity of the nearest-point projection on the tubular neighbourhood of sets with *positive reach*. However, attractors of dissipative PDEs generically have non-integer fractal dimension and reach zero (cf. [22], Chapter V). This rules out the direct application of Federer's theorem.

The purpose of this note is to introduce a weaker regularity class— *trajectory log-Lipschitz* projections—that is compatible with fractal geometry and still suffices for bounded-variation (BV) control of the projected trajectory. Motivated by log-Lipschitz regularity results for dissipative flows (cf. [4]), we observe that the naive global log-Lipschitz condition

$$\|P(x)-P(y)\| \;\leq\; C\,\|x-y\|\left(1+\log\tfrac{R}{\|x-y\|}\right) \tag{1}$$

does *not* imply BV of $P\circ u$ for AC curves u: the partition sums $\sum_i \Delta_i(1+\log(R/\Delta_i))$ diverge logarithmically as the mesh refines (Section 3). The correct formulation replaces the increment-scale weight $\log(R/\|x-y\|)$ with the attractor-distance weight $\log(R/d(t))$, where $d(t) = \operatorname{dist}(u(t),\mathcal{A})$. This yields a trajectory-adapted condition whose BV sums converge whenever a log-distance integrability condition (LDI) holds.

Main contributions. This paper makes three contributions: (i) We introduce the *trajectory log-Lipschitz* (TLL) condition and the *log-distance integrability* (LDI) condition as a trajectory-adapted regularity scale for BV projections onto fractal attractors, weaker than global Lipschitz projection regularity but stronger than a purely Hölder-type heuristic. (ii) We prove a BV chain rule (Theorem 6.1) showing that TLL + LDI suffices for bounded-variation selection regularity. (iii) We identify sufficient geometric conditions (bounded variation of the log-distance and sublevel-time control) that are compatible with the fractal geometry of dissipative PDE attractors.

Remark 1.1 (Input-output contract). The central result of this paper can be stated as an input-output contract:

Input: An absolutely continuous trajectory $u \in AC([0,T];H)$, a compact attractor $\mathcal{A}\subset H$, and a fixed measurable nearest-point selection $P : \mathcal{A}^R \to \mathcal{A}$.

Conditions: (TLL) The selected map P is trajectory log-Lipschitz along u; (LDI) the log-distance weight $\log(R/d(t))$ is $\|u'\|$-integrable.

Output: The composed selection $v(t) := P(u(t))$ belongs to $BV([0,T];H)$ with the explicit bound $\operatorname{Var}(v) \leq C_{\mathrm{TLL}} \int_0^T \|u'(t)\|\,\Omega(t)\,dt$.

This contract is designed to interface directly with the companion paper [13]: the output (BV selection) is precisely Assumption G' of that paper.

Assumption G' (from [13]): *There exists a nearest-point selection* $v(t) = P(u(t)) \in \mathcal{A}$ *such that* $v \in BV([0,T];\, H)$ *for every* $T < T^*$.

Combined with Condition D and the remaining hypotheses of [13] (including the energy-type inequality for the attractor distance), Assumption G' is the BV-selection input in that paper's conditional regularity argument. The present paper proves only the TLL + LDI $\Rightarrow$ BV step. The attractor $\mathcal{A}$ may have fractal geometry—TLL and LDI are compatible with non-integer Hausdorff dimension.

2 Setup and Notation

Throughout, H denotes a separable real Hilbert space with norm $\|\cdot\|$ and inner product $\langle \cdot, \cdot \rangle$.

Definition 2.1 (Dissipative semiflow and attractor)**.** A *dissipative semiflow* on H is a family $\{S(t)\}_{t \geq 0}$ of continuous maps $S(t) : H \to H$ satisfying $S(0) = \mathrm{id}$, $S(t+s) = S(t) \circ S(s)$, and possessing a *compact global attractor* $\mathcal{A} \subset H$: a compact, invariant ($S(t)\mathcal{A} = \mathcal{A}$ for all $t \geq 0$) set that attracts every bounded subset of H.

Definition 2.2 (Tubular neighbourhood and distance)**.** For $R > 0$, the *R-tubular neighbourhood* of $\mathcal{A}$ is

$$\mathcal{A}^R \;:=\; \{x \in H : \mathrm{dist}(x, \mathcal{A}) < R\},$$

where $\mathrm{dist}(x, \mathcal{A}) := \inf_{a \in \mathcal{A}} \|x - a\|$. For a curve $u : [0,T] \to \mathcal{A}^R$, we write $d(t) := \mathrm{dist}(u(t), \mathcal{A})$.

Definition 2.3 (Nearest-point selection)**.** A *nearest-point selection* is a measurable map $P : \mathcal{A}^R \to \mathcal{A}$ satisfying $\|x - P(x)\| = \mathrm{dist}(x, \mathcal{A})$ for all $x \in \mathcal{A}^R$, and $P|_{\mathcal{A}} = \mathrm{id}$. Such a selection exists by the Kuratowski–Ryll-Nardzewski theorem applied to the compact-valued multifunction $x \mapsto \arg\min_{a \in \mathcal{A}} \|x - a\|$.

Proposition 2.4 (Measurable and tie-broken nearest-point selections)**.** *Let* $u \in \mathrm{AC}([0,T];\, H)$ *and set*

$$\Pi(t) := \{a \in \mathcal{A} : \|u(t) - a\| = d(t)\}.$$

Then $\Pi(t)$ *is a non-empty compact-valued measurable multifunction, and hence admits measurable selections* $v(t) \in \Pi(t)$*. If a Borel tie-breaker* $R : \mathcal{A} \to \mathbb{R}$ *is fixed and attains a unique minimum on* $\Pi(t)$ *for a.e.* t *(for instance after a finite-dimensional or Moreau–Yosida regularisation), then*

$$v_R(t) \in \arg\min_{a \in \Pi(t)} R(a)$$

defines a canonical tie-broken nearest-point selection on those times. On the remaining ambiguous set, any measurable selection among the minimisers may be used.

Proof. Compactness and non-emptiness of $\Pi(t)$ follow from compactness of $\mathcal{A}$ and continuity of $a \mapsto \|u(t) - a\|$. The graph of Π is measurable because u is measurable and the distance function is continuous. The Kuratowski–Ryll-Nardzewski selection theorem therefore gives a measurable nearest-point selection. For a fixed Borel tie-breaker R, the restricted argmin correspondence $t \mapsto \arg\min_{\Pi(t)} R$ is again compact-valued and measurable; if the minimiser is unique for a.e. t, the stated tie-broken selection follows. The nearest-point literature records both ambiguous-locus phenomena in Banach spaces [7] and near-everywhere single-valuedness under additional hypotheses [23]; this is why the selector is fixed explicitly. No a.e. uniqueness of nearest points is asserted in general. □

Remark 2.5 (Role of tie-breaking). The BV chain rule below does not prove that a preferred nearest-point selection satisfies TLL. Rather, it assumes a measurable selection P with TLL and derives the BV conclusion. Tie-breakers such as minimal enstrophy, lexicographic finite-mode order, or a regularised metric-derivative rule are useful candidates for constructing such a selection, but their TLL regularity is an additional theorem to be proved in each concrete system.

Remark 2.6 (Uniqueness for non-convex sets). For non-convex $\mathcal{A}$, the nearest-point set $\Pi(t)$ may contain more than one element, and neither strict convexity of the Hilbert norm nor differentiability of the one-dimensional function $t \mapsto d(t)$ forces uniqueness of the nearest point in a non-convex set. Porosity results for the ambient ambiguity set are therefore best used as generic background [23, 7], not as an a.e. uniqueness claim along every absolutely continuous trajectory. This is why the main theorem is formulated for a fixed selection satisfying TLL.

3 Why Global Log-Lipschitz Fails

We first show that the global log-Lipschitz condition (1), even combined with log-distance integrability, does not directly yield BV of $P \circ u$.

Proposition 3.1 (Failure of the global LL bound to control variation)**.** *Let $P : \mathcal{A}^R \to \mathcal{A}$ satisfy* (1) *with constant $C > 0$. Let $u : [0, 1] \to \mathcal{A}^R$ be Lipschitz with $\|u'(t)\| = 1$ a.e. For the uniform partition $t_i = i/n$ $(i = 0, \dots, n)$, the upper bound obtained from* (1) *satisfies*

$$\sum_{i=0}^{n-1} \|P(u(t_{i+1})) - P(u(t_i))\| \;\le\; C\,(1 + \log(nR)),$$

which diverges as $n \to \infty$. In particular, the global LL condition does not provide a finite a priori *bound on* $\mathrm{Var}(P \circ u)$.

Proof. Each increment satisfies $\Delta_i := \|u(t_{i+1}) - u(t_i)\| = 1/n$. By (1):

$$\sum_{i=0}^{n-1} \|P(u(t_{i+1})) - P(u(t_i))\| \;\le\; C \sum_{i=0}^{n-1} \frac{1}{n} \Big(1 + \log(nR)\Big) \;=\; C\,(1 + \log(nR)).$$

As $n \to \infty$, this grows as $C \log n \to \infty$. Since the bound $C(1 + \log(nR))$ holds for *every* partition with mesh $1/n$, the supremum over all such partitions diverges: $\sup_n C(1 + \log(nR)) = \infty$. Hence the global LL condition does not provide a finite *a priori* bound on $\mathrm{Var}(P \circ u)$ via the standard estimate. (Whether the variation itself is infinite depends on the specific geometry of $\mathcal{A}$; the point is that the global LL condition is insufficient as a *tool* for establishing BV—no matter how the partition is chosen, the LL-derived upper bound diverges with refinement.) □

Remark 3.2. The divergence is *logarithmic*, not polynomial—much slower than for Hölder maps (where $\sum \Delta_i^\alpha$ converges for $\alpha < 1$). This makes global LL "almost" sufficient for BV, but the logarithmic accumulation over infinitely many partition points defeats it. The trajectory log-Lipschitz condition avoids this by anchoring the log weight at the attractor distance $d(t)$, which does not depend on the partition.

4 The Trajectory Log-Lipschitz Condition

Definition 4.1 (Trajectory log-Lipschitz projection (TLL))**.** Let $u \in \mathrm{AC}([0,T]; H)$ with $u(t) \in \mathcal{A}^R$ for all t, and let $P : \mathcal{A}^R \to \mathcal{A}$ be a nearest-point selection. We say that P satisfies the *trajectory log-Lipschitz condition* (TLL) along u if there exist constants $C_{\mathrm{TLL}} > 0$ and $h_0 > 0$ such that for all $t \in [0,T]$ with $d(t) > 0$ and all $h \in \mathbb{R}$ with $0 < |h| < h_0$ and $t + h \in [0,T]$ (i.e., both forward and backward increments):

$$\|P(u(t+h)) - P(u(t))\| \;\le\; C_{\mathrm{TLL}}\,\|u(t+h) - u(t)\|\,\Omega(t), \tag{2}$$

where $\Omega(t) := 1 + \log(R/d(t))$ is the *log-distance weight.*

On the set $\{t : d(t) = 0\}$ (where $u(t) \in \mathcal{A}$), $P(u(t)) = u(t)$ by idempotence of P. For the transition $d(t) = 0$, $d(t+h) > 0$, we have $\|P(u(t+h)) - u(t)\| = \|P(u(t+h)) - P(u(t))\|$. Since $u(t) \in \mathcal{A}$ is a candidate for the nearest point of $u(t+h)$, the minimality of P gives $\|u(t+h) - P(u(t+h))\| \le \|u(t+h) - u(t)\|$. Together with the triangle inequality:

$$\|P(u(t+h)) - u(t)\| \;\le\; \|P(u(t+h)) - u(t+h)\| + \|u(t+h) - u(t)\| \;\le\; 2\,\|u(t+h) - u(t)\|.$$

This gives $|v'|(t) \le 2\,|u'|(t)$ on $\{d = 0\}$.

Remark 4.2 (Convention $C_{\mathrm{TLL}} \ge 2$). The factor 2 on $\{d = 0\}$ is absorbed by the TLL bound provided $C_{\mathrm{TLL}} \ge 2$ and $\Omega(t) \ge 1$. Throughout, we adopt the convention $C_{\mathrm{TLL}} := \max(C_{\mathrm{TLL}}, 2)$; this changes the BV bound (4) by at most a factor of 2 and does not affect the integrability of (3).

Remark 4.3 (Geometric interpretation of TLL). The condition TLL asserts that the nearest-point projection, evaluated along the trajectory u, has a local Lipschitz constant that blows up *at most logarithmically* as the trajectory approaches the attractor:

$$\mathrm{Lip}_{\mathrm{loc}}(P \circ u;\, t) \;\lesssim\; 1 + \log \frac{R}{d(t)}.$$

At distance $d(t) \sim R$ from the attractor, the projection is essentially Lipschitz (the log term is $O(1)$). As $d(t) \to 0$, the log weight grows, reflecting the increasing geometric complexity of the attractor boundary at fine scales. The logarithmic rate is the *critical threshold*: faster growth (e.g., polynomial) would make LDI fail generically, while slower growth (bounded) would imply Lipschitz projection—too strong for fractal geometry.

Remark 4.4 (TLL as Aubin property of the projection correspondence). The TLL condition admits a natural reformulation in the language of variational analysis. Define the *projection correspondence*

$$\Gamma : t \;\mapsto\; \Big\{a \in \mathcal{A} : \|a - u(t)\| = d(t)\Big\}$$

mapping each time to the set of nearest attractor elements. The TLL condition (Definition 4.1) is *analogous* to the *Aubin property* (metric regularity) of the correspondence Γ at rate $1/|\log r|$. Specifically, Γ is metrically regular at rate $\phi(r) = 1/|\log r|$ if

$$\mathrm{dist}\Big(a,\, \Gamma(t')\Big) \;\le\; L\,\phi\Big(\|u(t') - u(t)\|\Big)\,\|u(t') - u(t)\|$$

for $a \in \Gamma(t)$ and t' near t. Note that TLL and the Aubin property are *not* equivalent in general: TLL weights by $\Omega(t) = 1 + \log(R/d(t))$ (the *attractor distance*), while the Aubin formulation weights by $1/|\log \|u(t') - u(t)\||$ (the *increment size*). The two formulations coincide in the special case where $d(t) \sim \|u(t') - u(t)\|$ (i.e., the trajectory increment and the attractor distance are of the same order), which occurs for trajectories approaching the attractor non-tangentially. This analogy connects the projection regularity theory to the Aubin criterion and metric regularity framework of Rockafellar–Wets [19] and Dontchev–Rockafellar [8]. The key advantage is that Aubin-property criteria are well-developed for set-valued maps and provide verifiable sufficient conditions (e.g., via the Mordukhovich coderivative) that may be applicable to the attractor projection.

Remark 4.5 (TLL vs global LL). The global log-Lipschitz condition (1) does *not* imply TLL. For $h \to 0$ at fixed t:

$$\log \frac{R}{\|u(t+h) - u(t)\|} \;\approx\; \log \frac{R}{h\,|u'(t)|} \;=\; \log \frac{R}{|u'(t)|} + \log \frac{1}{h} \;\to\; \infty,$$

while $\log(R/d(t))$ remains fixed. The global condition produces an *increment-scale* log weight that diverges with the partition mesh; TLL produces a *distance-scale* log weight that depends only on the current geometry. This is the structural reason why TLL yields BV while global LL does not.

5 Log-Distance Integrability

Definition 5.1 (Log-distance integrability (LDI))**.** In the setting of Definition 4.1, we say that *log-distance integrability* holds if

$$\int_{\{d>0\}} \|u'(t)\|\,\Omega(t)\,\mathrm{d}t \;=\; \int_{\{d>0\}} \|u'(t)\|\,\Big(1 + \log \tfrac{R}{d(t)}\Big)\,\mathrm{d}t \;<\; \infty. \tag{3}$$

On the set $\{t : d(t) = 0\}$, $v(t) = u(t)$ and the BV contribution is controlled by $\int_{\{d=0\}} \|u'(t)\|\,\mathrm{d}t < \infty$ (since u is AC), so only the region $\{d > 0\}$ is relevant for (3). To state the BV bound (4) uniformly over all of $[0,T]$, we extend Ω to $\{d = 0\}$ by setting $\Omega(t) := 1$ there (consistent with the factor-2 bound of Remark 4.2).

Remark 5.2 (Minimality of LDI). The condition LDI is positioned precisely between the known regularity classes:

- **Lipschitz projection** ($\Omega \equiv 1$): LDI reduces to $\int \|u'\| < \infty$, which is automatic for AC curves. This is the Federer/positive-reach regime.
- **Hölder projection** ($\Omega(t) \sim d(t)^{-\beta}$, $\beta > 0$): LDI requires $\int \|u'\|\, d(t)^{-\beta} < \infty$, which generically fails near the attractor. This is why Mané projectors (Hölder inverse) do not yield BV.
- **Log-Lipschitz projection** (TLL): LDI requires $\int \|u'\| \log(1/d) < \infty$, which is the borderline case—logarithmic singularities are integrable under mild dynamical conditions.

The log-Lipschitz/LDI pair thus occupies the *critical* regularity window for BV selections on fractal attractors.

6 Main Result: BV Chain Rule

Theorem 6.1 (BV chain rule for log-Lipschitz projections)**.** *Let $\{S(t)\}_{t\geq 0}$ be a dissipative semiflow on a separable Hilbert space H with compact global attractor $\mathcal{A} \subset H$. Let $u \in \mathrm{AC}([0,T];H)$ with $u(t) \in \mathcal{A}^R$ for all t, and let $P : \mathcal{A}^R \to \mathcal{A}$ be a nearest-point selection satisfying TLL (2) along u. Set $v(t) := P(u(t))$.*

If LDI (3) holds, then $v \in \mathrm{BV}([0,T];H)$ with the explicit bound

$$\mathrm{Var}_H(v;\,[0,T]) \;\leq\; C_{\mathrm{TLL}} \int_0^T \|u'(t)\|\,\Omega(t)\,\mathrm{d}t, \tag{4}$$

where $\Omega(t) := 1 + \log(R/d(t))$ on $\{d > 0\}$ and $\Omega(t) := 1$ on $\{d = 0\}$ (Remark 4.2).

Proof. The proof has three steps.

Step 1 (Metric derivative bound). For any continuous curve $v : [0,T] \to H$, the *upper metric derivative* at t is

$$|v'|^+(t) \;:=\; \limsup_{h\to 0} \frac{\|v(t+h) - v(t)\|}{|h|} \;\in\; [0,+\infty].$$

This quantity exists everywhere (as a value in $[0,+\infty]$) and is Borel measurable. For absolutely continuous curves the $\limsup$ is a genuine limit a.e. (cf. [2, Theorem 1.1.2]). Since u is AC, $|u'|(t) := \lim_{h\to 0} \|u(t+h) - u(t)\|/|h|$ exists for a.e. t.

Case $d(t) > 0$: Dividing (2) by $|h|$ and taking $\limsup$:

$$|v'|^+(t) \;\leq\; C_{\mathrm{TLL}}\,|u'|(t)\,\Omega(t) \qquad \text{for a.e. } t \text{ with } d(t) > 0. \tag{5}$$

(The right-hand side is a genuine limit; the left-hand side may be a $\limsup$, which is dominated by the same bound.)

Case $d(t) = 0$ (interior of $\mathcal{Z}$): On $\mathcal{Z} := \{t \in [0,T] : d(t) = 0\}$, $u(t) \in \mathcal{A}$ and $v(t) = P(u(t)) = u(t)$. Since $u(t) \in \mathcal{A}$ is a candidate minimiser for $\mathrm{dist}(u(t+h),\mathcal{A})$, the triangle inequality gives $\|P(u(t+h)) - u(t)\| \leq 2\,\|u(t+h) - u(t)\|$ (cf. the bound in Definition 4.1), so $|v'|^+(t) \leq 2\,|u'|(t)$.

Boundary points of $\mathcal{Z}$: At $t_0 \in \partial\mathcal{Z}$ with $d(t_0) = 0$, sequences $t_n \to t_0$ with $d(t_n) > 0$ may exist. For continuity of v at t_0: $\|v(t_n) - v(t_0)\| = \|P(u(t_n)) - u(t_0)\| \leq d(t_n) + \|u(t_n) - u(t_0)\| \to 0$, since u is continuous and $d = \mathrm{dist}(\cdot,\mathcal{A}) \circ u$ inherits continuity (the distance function is 1-Lipschitz). The metric derivative bound $|v'|^+(t_0) \leq 2\,|u'|(t_0)$ remains valid by the same triangle inequality argument as above.

Since $\Omega(t) \geq 1$ and $C_{\mathrm{TLL}} \geq 2$ (Remark 4.2), the bound (5) holds on all of $[0,T]$ (with $|v'|^+$ in place of $|v'|$ where the limit may not exist; this suffices for the BV criterion in Step 3).

Step 2 (Continuity of v). The nearest-point projection on a compact (but non-convex) set is in general only upper semicontinuous as a set-valued map and need not be single-valued everywhere. However, the TLL condition (2) directly implies continuity of $v = P \circ u$ along the trajectory: for $d(t) > 0$, $\|v(t+h) - v(t)\| \leq C_{\mathrm{TLL}}\,\|u(t+h) - u(t)\|\,\Omega(t) \to 0$ as $h \to 0$; for $d(t) = 0$, $\|v(t+h) - v(t)\| \leq 2\,\|u(t+h) - u(t)\| \to 0$ (including boundary points of $\mathcal{Z}$, cf. Step 1). Hence v is continuous on $[0,T]$.

Step 3 (BV bound). We show $\|v(b) - v(a)\| \leq \int_a^b g(t)\,\mathrm{d}t$ for all $0 \leq a < b \leq T$, where $g(t) := C_{\mathrm{TLL}}\,|u'|(t)\,\Omega(t) \in L^1$ (by LDI). Since v is continuous (Step 2), the BV bound $\mathrm{Var}(v) \leq \int_0^T g$ then follows by summing over any partition.

Set $f(t) := \|v(t) - v(a)\|$. Then $f : [a, b] \to \mathbb{R}$ is continuous and real-valued, $f(a) = 0$, and the upper Dini derivative satisfies $D^+ f(t) \le |v'|^+(t)$ (since $f(t+h) - f(t) \le \|v(t+h) - v(t)\|$ by the reverse triangle inequality, and division by $h > 0$ followed by lim sup gives $D^+ f \le |v'|^+$). By Step 1, $|v'|^+(t) \le g(t)$ a.e. Moreover, $f(t) = \|v(t) - v(a)\| \ge 0$, so $D_+ f(t) \ge -|v'|^+(t) > -\infty$ a.e. (since $g \in L^1$ implies $|v'|^+ < \infty$ a.e.). A classical result on Dini derivatives (see, e.g., [21, Chapter VII, §§ 9–10]; cf. also [1, Chapter 3]) gives: for a continuous function $f : [a, b] \to \mathbb{R}$ with $D^+ f(t) \le g(t)$ a.e., $D_+ f(t) > -\infty$ a.e., and $g \in L^1$, the inequality $f(b) - f(a) \le \int_a^b g(t)\,\mathrm{d}t$ holds. Therefore

$$\|v(b) - v(a)\| \;=\; f(b) \;\le\; \int_a^b g(t)\,\mathrm{d}t \;\le\; \int_0^T g(t)\,\mathrm{d}t.$$

Summing over any partition $0 = t_0 < \cdots < t_n = T$:

$$\sum_i \|v(t_{i+1}) - v(t_i)\| \;\le\; \sum_i \int_{t_i}^{t_{i+1}} g(t)\,\mathrm{d}t \;=\; \int_0^T g(t)\,\mathrm{d}t \;=\; C_{\mathrm{TLL}} \int_0^T |u'|(t)\,\Omega(t)\,\mathrm{d}t,$$

which gives (4).

The bound (4) follows from Steps 1–3. ☐

Remark 6.2 (Direct partition-sum verification). For a fixed partition with mesh $< h_0$, the finiteness of the partition sum can be seen directly from TLL without the Dini-derivative machinery: on each subinterval, TLL gives $\|v(t_{i+1}) - v(t_i)\| \le C_{\mathrm{TLL}}\, \|u(t_{i+1}) - u(t_i)\|\, \Omega(t_i)$ (or the factor-2 bound on $\{d = 0\}$, absorbed by $C_{\mathrm{TLL}} \ge 2$). The resulting partition sum $\sum_i C_{\mathrm{TLL}}\, \Omega(t_i) \int_{t_i}^{t_{i+1}} \|u'\|\,\mathrm{d}t$ is finite for each fixed partition (since Ω is bounded on any compact set disjoint from $\{d = 0\}$, and the $\{d = 0\}$ contribution is controlled by $2 \int \|u'\|$). However, the passage to the supremum over *all* partitions—which is the definition of $\mathrm{Var}(v)$—requires the Dini-derivative argument of Step 3, which sharpens the pointwise $\Omega(t_i)$-weight to the integrated $\Omega(t)$-weight.

Remark 6.3 (Banach space generality). The proof of Theorem 6.1 uses only the norm, compactness of $\mathcal{A}$, and measurable selection—no inner product or Hilbert space structure. The BV chain rule therefore holds verbatim in any separable Banach space. We restrict to Hilbert spaces only for the PDE applications in Section 8.

7 Sufficient Conditions for LDI

The condition LDI (3) is a single integrability requirement. We provide two independent sufficient conditions, each verifiable from dynamical information about the semiflow.

7.1 Variant A: Log-distance has bounded variation

Proposition 7.1 (BV log-distance implies LDI)**.** *Suppose $d(t) > 0$ for a.e. $t \in [0, T]$ and $\log(R/d(t)) \in \mathrm{BV}([0, T])$. Then LDI holds.*

Proof. Write $\ell(t) := 1 + \log(R/d(t))$. Since $u' \in L^1(0, T; H)$ (as u is AC) and $\ell \in \mathrm{BV}([0, T]) \subset L^\infty([0, T])$, the product $\|u'(t)\|\, \ell(t)$ is locally integrable:

$$\int_0^T \|u'(t)\|\, \ell(t)\,\mathrm{d}t \;\le\; \|\ell\|_{L^\infty(0,T)} \int_0^T \|u'(t)\|\,\mathrm{d}t \;<\; \infty. \qquad ☐$$

Remark 7.2 (When is log-distance BV?). The condition $\log(R/d) \in \mathrm{BV}$ requires that the trajectory does not oscillate infinitely often between different distance scales from the attractor. Formally, it excludes infinite "zickzack frequency" in $d(t)$. For monotonically attracting trajectories ($d(t)$ eventually decreasing), this is automatic. For trajectories with transient oscillations, BV of $\log d$ follows if the total number of scale crossings $\{t : d(t) = \varepsilon\}$ is finite for each $\varepsilon > 0$—a generic property of dissipative semiflows with isolated equilibria.

7.2 Variant B: Sublevel time-measure control

Definition 7.3 (Sublevel-time control (STC))**.** The trajectory u satisfies *sublevel-time control* with exponent $\alpha > 0$ if there exists $C_* > 0$ such that

$$\Big|\{t \in [0,T] : d(t) \le \varepsilon\}\Big| \;\le\; C_*\,\varepsilon^{\alpha} \qquad \text{for all } \varepsilon \in (0,R]. \tag{6}$$

Remark 7.4 (STC implies $|\{d = 0\}| = 0$). Taking $\varepsilon \to 0$ in (6) shows $|\{d = 0\}| = 0$. Hence STC is only applicable to trajectories that do not spend positive time on the attractor. This is the generic case for non-stationary solutions approaching $\mathcal{A}$ asymptotically. For trajectories that reach the attractor in finite time (e.g., convergence to an equilibrium with $u(t) \in \mathcal{A}$ for $t \ge t_1$), Variant A (BV log-distance) is the appropriate sufficient condition.

Proposition 7.5 (STC implies LDI)**.** *If STC* (6) *holds with some* $\alpha > 0$ *and* $u' \in L^2(0,T;\,H)$*, then* $\log(R/d) \in L^p(0,T)$ *for every* $p < \infty$*, and LDI holds.*

Proof. **Step 1 (Tonelli representation).** Since $u(t) \in \mathcal{A}^R$, we have $d(t) \le R$ for all t. By Remark 7.4, STC implies $|\{d = 0\}| = 0$. All logarithmic integrals below are therefore taken over $\{d > 0\}$; trajectories with positive-measure contact with the attractor are handled by Variant A rather than by STC. On $\{t : d(t) > 0\}$, using the layer-cake identity $\log(R/d(t)) = \int_{d(t)}^{R} s^{-1}\,\mathrm{d}s$ and Tonelli:

$$\int_{\{d>0\}} \|u'(t)\|\,\log\frac{R}{d(t)}\,\mathrm{d}t \;=\; \int_0^R \frac{1}{s}\Big(\int_{\{0<d(t)\le s\}} \|u'(t)\|\,\mathrm{d}t\Big)\,\mathrm{d}s. \tag{7}$$

Step 2 (Hölder on inner integral). By Hölder's inequality:

$$\int_{\{0<d(t)\le s\}} \|u'(t)\|\,\mathrm{d}t \;\le\; \Big(\int_0^T \|u'(t)\|^2\,\mathrm{d}t\Big)^{1/2} \cdot \Big|\{0 < d \le s\}\Big|^{1/2} \;\le\; \|u'\|_{L^2}\,C_*^{1/2}\,s^{\alpha/2},$$

where the last step uses $|\{0 < d \le s\}| \le |\{d \le s\}| \le C_*\,s^{\alpha}$ from STC.

Step 3 (Integrability near $s = 0$). Substituting into (7):

$$\int_0^R \frac{1}{s}\,\|u'\|_{L^2}\,C_*^{1/2}\,s^{\alpha/2}\,\mathrm{d}s \;=\; \|u'\|_{L^2}\,C_*^{1/2}\int_0^R s^{\alpha/2-1}\,\mathrm{d}s \;=\; \|u'\|_{L^2}\,C_*^{1/2}\,\frac{2}{\alpha}\,R^{\alpha/2} \;<\; \infty.$$

The integral converges because $\alpha > 0$, so $s^{\alpha/2-1}$ is integrable near $s = 0$.

Step 4 (L^p bounds). The same Tonelli argument with Hölder exponents p and q ($1/p + 1/q = 1$) gives $\|\log(R/d)\|_{L^p} \le C(p,\alpha,R,C_*)$ for every $p < \infty$, so LDI follows from any finite L^q bound on $\|u'\|$. □

Remark 7.6 (Regularity of u' for Navier–Stokes). Proposition 7.5 requires $u' \in L^2(0,T;\, H)$. For the 3D Navier–Stokes equations, this is not a consequence of the energy-class bounds $u \in L^2(0,T;\, V) \cap L^\infty(0,T;\, H)$. One obtains this H-valued time derivative only under stronger strong-solution hypotheses, for instance $u \in L^\infty(0,T;\, V) \cap L^2(0,T;\, D(A))$, together with standard assumptions on the forcing and the equation. For *Leray–Hopf weak solutions*, one only has $u' \in L^{4/3}(0,T;\, V')$, where $V' = H^{-1}$. In that setting, the Hölder step (Step 2 of the proof) must be adapted to use the V'-norm, with the STC exponent adjusted accordingly, or Proposition 7.5 must be used only inside the stronger solution class. This is a genuine transfer gate, not an automatic Leray–Hopf input.

Remark 7.7 (Sources of STC in dissipative flows). Sublevel-time control (6) is the genuine new analytical input. Three mechanisms can generate it:

(i) **Finite-window lower distance envelope:** A one-sided attraction estimate $d(t) \le C_0 e^{-\lambda t}$ does not control $\log(R/d(t))$ from above and does not by itself imply STC. What is needed is a non-superexponential lower envelope such as $d(t) \ge c_T e^{-\lambda_T t}$ on a fixed time window $[0,T]$. Then $\{d \le \varepsilon\}$ can occur only after the threshold $(\log(c_T/\varepsilon))/\lambda_T$, giving finite-window STC constants (depending on T, c_T, λ_T) and, more directly, the logarithmic bound in Lemma 7.8.

(ii) **Lyapunov dissipation:** If a Lyapunov functional E satisfies $-\dot{E} \ge \Psi(d)$ with Ψ not too flat near $d = 0$ (e.g. $\Psi(d) \ge c\, d^\beta$ for some β), then sublevel times are controlled by the total energy budget $E(0) - \inf E$.

(iii) **No fast oscillation:** Energetic constraints (finite dissipation integral) prevent arbitrarily rapid back-and-forth motion near the attractor, bounding the number of excursions into $\{d \le \varepsilon\}$.

Each mechanism yields STC with different exponents α. The STC condition is a plausible candidate hypothesis in many standard dissipative PDEs, but it is not automatic and must be proved or measured in each concrete system.

Lemma 7.8 (Exponential lower envelope implies log-integrability)**.** *Let* $0 < c \le R$*,* $\lambda \ge 0$*, and suppose that on* $[0,T]$ *the trajectory satisfies*

$$0 < d(t) \le R, \qquad d(t) \ge c\, e^{-\lambda t}.$$

Then $\log(R/d) \in L^\infty(0,T)$*, and*

$$\int_0^T \log \frac{R}{d(t)}\, \mathrm{d}t \;\le\; T \log \frac{R}{c} + \frac{\lambda T^2}{2}.$$

If additionally $u' \in L^1(0,T;\, H)$*, then LDI holds on* $[0,T]$*.*

Proof. The lower envelope gives

$$\log \frac{R}{d(t)} \;\le\; \log \frac{R}{c\, e^{-\lambda t}} \;=\; \log \frac{R}{c} + \lambda t.$$

Hence

$$\int_0^T \log \frac{R}{d(t)}\, \mathrm{d}t \;\le\; \int_0^T \Big(\log(R/c) + \lambda t\Big)\, \mathrm{d}t \;=\; T \log \frac{R}{c} + \frac{\lambda T^2}{2}.$$

For the LDI condition (3):

$$\int_0^T \|u'(t)\| \log \frac{R}{d(t)}\,\mathrm{d}t \;\le\; \left(\log \frac{R}{c} + \lambda T\right)\|u'\|_{L^1(0,T;H)} \;<\; \infty$$

because the logarithmic factor is bounded on the finite window. □

Remark 7.9 (Exponential envelopes and STC). Lemma 7.8 is deliberately stated with a lower envelope. A pure attraction estimate $d(t) \le C_0 e^{-\lambda t}$ gives only an upper bound on the distance to the attractor; it does not prevent $d(t)$ from being much smaller and therefore cannot bound $\log(R/d(t))$ from above. Thus exponential attraction is useful evidence, but LDI requires either direct STC, BV control of the log-distance, or a lower envelope/no-superexponential-approach estimate.

Remark 7.10 (STC as the realistic candidate for Navier–Stokes). Of the three sufficient conditions for LDI (Variant A: BV log-distance; Variant B: sublevel time control; Variant C: a lower exponential distance envelope), *sublevel time control* (STC) is the most realistic candidate for 3D Navier–Stokes:

1. *Exponential attraction* ($d(t) \le C_0 e^{-\lambda t}$) is the easy dynamical regime far from blow-up, but by itself it is not an LDI proof. LDI uses either direct STC or a lower distance envelope as in Lemma 7.8.
2. *STC* ($|\{d(t) \le \varepsilon\}| \le C_* \varepsilon^{\alpha}$) is robust: it only requires that the trajectory does not spend too long near the attractor at each scale. Two PDE mechanisms make STC plausible for Navier–Stokes:
   - *Dissipative time-scale separation:* Near the attractor, the low-mode dynamics are slow (controlled by the attractor) while the high-mode dynamics are fast (viscous damping). This separation limits the residence time at any given distance scale.
   - *Finite determining modes:* The number of determining modes $N_d \sim G^{2/3}$ (where G is the Grashof number) limits the effective dimension of the dynamics near $\mathcal{A}$, preventing pathological accumulation at any distance scale.
3. *BV log-distance* (Variant A) requires monotone approach to the attractor, which is generically violated by oscillatory solutions.

8 Applications to Dissipative PDEs

We illustrate how the abstract framework applies to concrete PDE settings. In systems with a genuine single-valued semiflow and a compact global attractor, the question is whether TLL and LDI hold. For three-dimensional Navier–Stokes one must distinguish this conditional strong-semiflow setting from the unconditional Leray–Hopf/trajectory-attractor setting.

8.1 3D Navier–Stokes on $\mathbb{T}^3$

Example 8.1 (Navier–Stokes equations). Let $H = L^2(\mathbb{T}^3; \mathbb{R}^3)$ (divergence-free) and let f be smooth. The following application is conditional on working in a strong single-valued semiflow, or in a selected trajectory-attractor framework where the same compactness and selection objects are fixed in advance. In the unconditional Leray–Hopf theory, non-uniqueness prevents one from simply invoking a Temam-style global strong attractor as an ordinary semiflow object. In the conditional strong-attractor regime, the standard attractor theory gives:

- $\mathcal{A}$ is compact in H^1 (hence in H).
- Every $a \in \mathcal{A}$ is C^∞.
- Exponential tracking holds *conditional on global regularity*: for strong solutions existing on $[0, \infty)$, $d(t) \le C_0\, e^{-\lambda t}$ for $t \ge t_0$. (For Leray–Hopf weak solutions, only algebraic attraction in L^2 is known unconditionally.) This attraction estimate is not by itself an LDI proof; it must be paired with STC, BV log-distance control, or a lower envelope.

In regimes where the finite-dimensional attractor estimates apply, the attractor has finite fractal dimension $d_f(\mathcal{A}) \le c_1\, G^{2/3}(1 + \log G)^{1/3}$ (where G is the Grashof number; cf. [6]), but generically has reach 0—ruling out Federer's theorem.

Status of TLL: Open. In a strong or selected Navier–Stokes semiflow, the squeezing property (cf. [11], Chapter 14) provides exponential contraction of high-frequency differences on the attractor. Combined with Mané projector estimates, this suggests a possible finite-mode route to TLL, but it does not by itself control the nearest-point map. A rigorous proof requires quantitative control of the projection stability in the high-mode complement $Q_N H$ and of the selected nearest-point branch.

Status of LDI: Open for the critical 3D problem. It follows on fixed windows from STC or from the lower-envelope criterion of Lemma 7.8, but not from the one-sided attraction estimate $d(t) \le C_0 e^{-\lambda t}$ alone. The critical question is the blow-up regime $T \nearrow T^*$, where $\|u'\| \to \infty$ and the distance geometry may degenerate. If TLL and LDI are established in a non-circular selected framework, the BV bound (4) yields Assumption G′ of [13]; the companion paper then uses this as part of its conditional attractor-distance regularity argument.

Connection to the log-Dirichlet quotient: The companion paper [13] introduces the *log-Dirichlet quotient* $Q_{\mathrm{LD}}(t) = \|\nabla(u - v)\|^2/\|u - v\|^2$ and shows that its integrability transforms the Gronwall estimate from exponential to algebraic growth. The LDI condition of the present paper supplies the selection-variation input for this route: the BV chain rule controls the relevant log-distance variation. An actual Q_{LD} bound still requires the comparability and energy hypotheses stated in the companion framework.

8.2 Reaction-diffusion systems

Example 8.2 (Reaction-diffusion on bounded domains). Consider $\partial_t u = D\Delta u + f(u)$ on a bounded domain $\Omega \subset \mathbb{R}^n$ with Dirichlet or Neumann boundary conditions, where $D > 0$ is a diffusion matrix and f satisfies a dissipativity condition $\langle f(u), u\rangle \le C - c|u|^2$. The semiflow possesses a compact global attractor $\mathcal{A} \subset L^2(\Omega)$ with finite fractal dimension [22].

For scalar equations ($u \in \mathbb{R}$) in one space dimension with a gradient nonlinearity $f(u) = -V'(u)$, the attractor consists of equilibria and heteroclinic orbits. In regimes where the inertial-manifold branch is sufficiently regular (for instance prox-regular or $C^{1,1}$ in a tubular neighbourhood), the nearest-point projection is Lipschitz on that branch; then

TLL and LDI follow in the sense of Theorem 13.5. Mallet-Paret–Sell type inertial-manifold results provide the natural test setting [16], while sharp dimension estimates for inertial manifolds give a quantitative reference point for this route [20].

For systems in higher dimensions, the attractor may have non-integer fractal dimension and reach zero. The log-Lipschitz framework then provides a path to BV selections that Federer's theorem does not. The spectral gap condition and sharp inertial-manifold dimension estimates are available in many reaction-diffusion settings [9], making the attractor geometry more regular than in the Navier–Stokes case.

8.3 Damped wave equations

Example 8.3 (Damped semilinear wave equation). Consider $\partial_{tt}u + \gamma\,\partial_t u = \Delta u + f(u)$ on $\Omega \subset \mathbb{R}^3$ with $\gamma > 0$ and subcritical nonlinearity. The semiflow on $H_0^1 \times L^2$ possesses a compact global attractor with finite fractal dimension (cf. [3]). The exponential tracking property holds with rate $\lambda = \gamma/2$ (spectral gap of the damped wave operator; cf. [3], Chapter V). For LDI, this attraction estimate is still only a dynamical input; it must be combined with STC or a lower distance envelope as above.

The attractor geometry is intermediate between reaction-diffusion (where inertial manifolds often exist) and Navier–Stokes (where they do not). The log-Lipschitz framework provides a *uniform* approach applicable across all three settings, with TLL as the single geometric input.

9 Discussion

9.1 The new mathematical interface

The central contribution of this note is the identification of the *critical regularity interface* for BV selections on fractal attractors. The interface is characterised by a single question:

Core Question. For a dissipative semiflow with compact attractor $\mathcal{A} \subset H$, does the nearest-point projection satisfy trajectory log-Lipschitz control (2)?

This is a question about the *regularity of the attractor as seen from the trajectory*, not about the global geometry of $\mathcal{A}$. The distinction is crucial: global geometric conditions (positive reach, inertial manifold) impose structure on the *entire* attractor, while TLL requires regularity only in the direction of the dynamical approach.

9.2 Relation to existing regularity theories

Condition	Projection	Composition	Fractal?	Source
Positive reach > 0	Lipschitz-1	AC $\Rightarrow$ AC	No	Federer [10]
Prox-regular	locally Lipschitz	AC $\Rightarrow$ AC (local)	No	Poliquin et al. [18]
Hausdorff-BV set map	BV selection exists	—	Yes	Chistyakov [5]
Mané projector	Hölder^{-1}	AC $\Rightarrow$ Hölder	Yes	Mané [17]
TLL + LDI	**log-Lipschitz**	**AC $\Rightarrow$ BV**	**Yes**	**This note**

The TLL+LDI framework is the only entry that simultaneously (i) accommodates fractal geometry (non-integer dimension, reach 0), (ii) yields BV regularity (sufficient for Gronwall-type arguments), and (iii) is formulated as a trajectory-level condition (independent of global attractor structure).

9.3 Limitations

- **Selection dependence.** The TLL condition is formulated for a *given* measurable nearest-point selection P. On non-convex compact sets, the nearest-point map is generically multi-valued, and different selections may have different regularity. The framework requires the existence of *at least one* selection satisfying TLL; it does not claim that every measurable selection does so. In practice, pre-registered tie-broken selections (e.g., minimal enstrophy, finite-mode lexicographic order, or a regularised metric-derivative rule) are the natural candidates.

- **TLL is open for 3D NS.** The squeezing property and Mané projector estimates suggest log-Lipschitz stability of the projection, but a rigorous proof requires quantitative bounds on the high-mode projection complement that are not currently available.

- **LDI is non-trivial only under blow-up.** For smooth solutions on compact time intervals, LDI holds automatically. The condition becomes critical only in the limit $T \nearrow T^*$ (blow-up time), where $\|u'\| \to \infty$.

- **BV, not AC.** The framework yields BV regularity, which suffices for Stieltjes-integral-based Gronwall arguments but not for pointwise differential inequalities. Upgrading BV to AC would require Lipschitz projection (positive reach)—exactly the condition we are trying to avoid.

- **BV bound depends on $\|u'\|$.** The bound (4) is $\mathrm{Var}(v) \le C_{\mathrm{TLL}} \int_0^T \|u'\| \, \Omega \, dt$. For Navier–Stokes, $\|u'\|$ depends on the solution norm, and near a potential blow-up ($T \nearrow T^*$), $\|u'\| \to \infty$. The Gronwall closure requires that the BV bound can be "absorbed" before $\mathrm{Var}(v)$ diverges. The details of this absorption depend on the companion paper [13] and are beyond the scope of this note.

Central open problem

> **Selection Regularity Problem.** In a strong Navier–Stokes semiflow, or in a preselected trajectory-attractor formulation for Leray–Hopf solutions, does the nearest-point projection or selected nearest-point correspondence $P_{\mathcal{A}}$ satisfy the TLL condition along $u(t)$? Equivalently: does the attractor geometry permit non-circular BV-regular selections?
>
> An affirmative answer, combined with the BV chain rule of this paper, would supply Assumption G' to the companion framework [13]. Any global-regularity conclusion still also depends on Condition D and the remaining hypotheses of that conditional framework.

10 Conclusion

We have introduced a regularity framework for nearest-point projections onto compact attractors that occupies the critical gap between Hölder (insufficient for BV) and Lipschitz (incompatible with fractal geometry). The trajectory log-Lipschitz condition (TLL) controls the projection modulus at the attractor distance scale, and the log-distance integrability condition (LDI) ensures that the resulting BV sums converge.

The framework reduces conditional regularity criteria for dissipative PDEs—including strong or preselected three-dimensional Navier–Stokes formulations—to a single geometric-dynamical question: does a fixed nearest-point selection have at most logarithmic blowup along dissipative trajectories? This question is not a substitute for the PDE-specific compactness, selection, and limiting hypotheses; it isolates the geometric part of that burden.

11 Time-averaged selection and BV regularity (speculative)

This section is speculative: it outlines an alternative approach whose key steps remain open conjectures. The results of Sections 6–8 do not depend on this material.

The preferred transfer architecture is now the finite-mode skeleton limit with pre-registered regularised selectors (Section 12). The time-averaged construction below is retained only as an auxiliary selection heuristic.

The central obstruction to closing the regularity gap is the BV control of the nearest-point selection $v(t) \in \arg\min_{a\in\mathcal{A}} \|u(t) - a\|$. We propose replacing the pointwise selection by a *time-averaged* selection that may have better BV behaviour after additional geometric input.

Definition 11.1 (Moreau–Yosida regularised selection)**.** For $\varepsilon > 0$, define the *time-averaged minimiser set*

$$\Pi_\varepsilon(t) \ := \ \arg\min_{a\in\mathcal{A}} \int_{t-\varepsilon}^{t+\varepsilon} \|u(s) - a\|^2 \, ds.$$

Any selected element $v_\varepsilon(t) \in \Pi_\varepsilon(t)$ must be chosen by a pre-registered measurable tie-breaker.

Conjecture 11.2 (Properties of v_ε)**.** *Under the standing assumptions (finite-dimensional attractor, energy dissipation bound), a pre-registered regularised selection $v_\varepsilon(t) \in \Pi_\varepsilon(t)$ is a candidate for the following properties:*

(i) Uniqueness: *$v_\varepsilon(t)$ is unique for a.e. t (strict convexity of the time-integrated cost on H; for minimisation over the non-convex compact set $\mathcal{A}$, uniqueness requires a separate argument via Hausdorff stability of the minimiser set, which remains open).*

(ii) Fixed-ε BV bound: $\mathrm{Var}(v_\varepsilon; [0,T]) \ \le \ C_\varepsilon \int_0^T \|\partial_t u(s)\| \, ds \ < \ \infty$*, for a branch for which the sensitivity estimate below is valid, with C_ε typically blowing up as $\varepsilon \to 0$. A* uniform *BV bound (independent of ε) is an additional geometric hypothesis, not a consequence of the definition; see Remark 11.4.*

(iii) TLL inheritance: *If u satisfies TLL with constant C_{TLL}, then v_ε is expected to satisfy TLL with constant $C_{\mathrm{TLL}} + O(\varepsilon)$. A rigorous proof requires control of the mismatch between pointwise and averaged distance, which remains open.*

(iv) Gronwall compatibility: *The Gronwall estimates of the main theorem (Theorem 6.1) are expected to remain valid for v_ε with an additional error term $O(\varepsilon \|\partial_t u\|_{L^2})$. A rigorous formulation requires specifying the relative entropy functional H and the constants α, β, γ; this is deferred to future work.*

Remark 11.3 (Partial heuristic). *(i)* The map $a \mapsto \int_{t-\varepsilon}^{t+\varepsilon} \|u(s) - a\|^2 \, ds$ is strictly convex on H. However, the minimisation is over the compact, generally *non-convex* set $\mathcal{A}$. Uniqueness of the minimiser over a non-convex set requires a separate argument (e.g., via Hausdorff stability of the minimiser set or a generic transversality argument). This step remains open for general attractors; it holds when $\mathcal{A}$ is a Lipschitz manifold.

(ii) If a chosen branch is unique and locally stable under time shifts, the usual first-variation sensitivity estimate suggests

$$\|v_\varepsilon(t+h) - v_\varepsilon(t)\| \;\le\; \frac{1}{2\varepsilon} \int_{t-\varepsilon}^{t+\varepsilon} \|u(s+h) - u(s)\| \, ds \;\le\; \frac{h}{2\varepsilon} \int_{t-\varepsilon}^{t+\varepsilon} \|\partial_t u(s)\| \, ds.$$

Integrating over $t \in [0, T]$ and applying Fubini would give a fixed-ε BV bound with a constant of order ε^{-1}. This is not a proof for arbitrary non-convex attractors; it only records the estimate that must be justified for a stable selected branch.

(iii) The TLL constant degrades by the time-averaging window:

$$|\log d_\varepsilon(t+h) - \log d_\varepsilon(t)| \;\le\; C_{\mathrm{TLL}} \, |\log |h|| \;+\; C\,\varepsilon/|h|,$$

where the second term arises from the mismatch between pointwise and averaged distance.

(iv) The Gronwall estimate picks up an additional forcing term from the time-averaging:

$$\frac{d}{dt} H(u \mid v_\varepsilon) \;\le\; -\alpha\, H + \beta + \gamma\, \varepsilon,$$

where γ depends on $\|\partial_t u\|$ but not on H. The standard Gronwall lemma applies; the additional $\gamma\varepsilon$ term vanishes in the limit $\varepsilon \to 0$.

Remark 11.4 (Passage to the limit $\varepsilon \to 0$). The key question is whether the BV bound survives the limit $\varepsilon \to 0$:

- If $\sup_\varepsilon \mathrm{Var}(v_\varepsilon;\, [0, T]) < \infty$, then a subsequence v_{ε_k} converges in L^1 to a BV nearest-point selection v_0 determined by the chosen regularisation and tie-breaking scheme.
- A uniform BV bound would amount to a new geometric condition that one may call "time-averaged reach". This phrase is only a name for the missing stability hypothesis; no equivalence with a known reach notion is claimed here.

This is the precise formulation of the "BV regularity gap" and connects to:

- *Moreau–Yosida regularisation* in convex analysis (the time-averaged cost is the Moreau envelope),
- *Viscous selection* in gradient systems (the time average acts as artificial viscosity),
- *Thermodynamic selection* in the Dark Energy companion paper (the IR averaging of $\Phi[\rho]$).

12 Post-Zookeeper Transfer: From Proof of Life to Damped 3D Navier–Stokes

The Zookeeper normal form makes this paper the natural test carrier for the Navier–Stokes branch. Here the defect, gap, and selection mechanism are already explicit:

In the current companion-program reading, this statement means RFEP v1.8 imports a normal-form checklist from the Zookeeper closure of Riemann $C^{(2)}$/MS2. It is not yet a proof of three-dimensional Navier–Stokes regularity; the damped-to-classical transfer remains the branch-specific obligation.

Operator/form.
The distance-to-attractor map and its measurable nearest-point selections.

Defect.
The log-distance integral, projection variation, and BV defect.

Approximation.
Lorenz $\to$ Kuramoto–Sivashinsky $\to$ reaction-diffusion $\to$ damped 3D Navier–Stokes $\to$ classical 3D Navier–Stokes.

Gap.
TLL, squeezing, lower distance envelopes, and damping gaps.

Limit.
From finite-dimensional attractors to PDE attractors, and from damped/hyperviscous systems to the classical $\beta = 1$ equation.

The immediate next target is therefore not the full Millennium problem. It is a damped 3D Navier–Stokes transfer theorem: in a parameter regime where the dynamical-systems literature supplies a global attractor and improved regularity, one should show that damping plus independently controlled projection-branch geometry and LDI imply BV selection. The existing Lorenz and Kuramoto–Sivashinsky tests then become the first two rungs of a larger ladder rather than isolated numerical evidence.

A non-circular transfer should therefore be formulated as a skeleton compactness problem. For a regularised system let $A_{\delta,N} = P_N \mathcal{A}_\delta$ and define a pre-registered selector

$$v_{\delta,N,\eta}(t) \in \arg\min_{a\in A_{\delta,N}} \frac{1}{2}\|P_N u_\delta(t) - a\|^2 + \eta R_N(a),$$

where R_N is fixed before observing success, for instance finite-mode lexicographic order, minimal enstrophy, or minimal metric derivative. The required theorem is not that damped success implies classical success. It is that skeleton convergence, uniform skeleton-TLL, uniform tie/sublevel-time control, and uniform BV bounds imply a BV limit selection for the classical problem.

Remark 12.1 (Tide-clock and bad-channel ledger for skeleton limits). The skeleton compactness criterion also has to localise where the TLL/LDI/BV costs are paid. For a finite skeleton let $d_1(t)$ and $d_2(t)$ be the nearest and second-nearest skeleton distances and set $g_{\mathrm{tie}}(t) = d_2(t) - d_1(t)$. For $K(\varepsilon) = \{t : g_{\mathrm{tie}}(t) \le \varepsilon\}$, future reports should record not only global C_{TLL}, LDI/L^1, and BV ratio, but also the LDI share $\int_{K(\varepsilon)} \|u'\|\Omega / \int \|u'\|\Omega$, the BV share on the same window, switching share, high-TLL share, and second-branch variation stress.

A first KS smoke test at $N = 64$ and $T_{\mathrm{test}} = 50$ gives $C_{\mathrm{TLL}}(95/99/\max) = 1.45042/40.9229/58.03$, LDI/$L^1 = 1.00838$, and a fitted tide-clock exponent $\alpha_{\mathrm{tie}} = 1.14757$. The closest 1% tie window carries about 0.99% of LDI and 0.69% of BV, while the 5% and 10% windows carry about 4.72%/9.55% of LDI but 11.77%/22.04% of BV. Thus this smoke test does not reveal a hidden LDI tail in the smallest near-tie set, but it does show visible BV concentration in wider tie corridors. This is a diagnostic guardrail, not a Navier–Stokes proof. Consequently a future no-hidden-backflow lemma should require the tie-window LDI shares, BV shares, branch stress, and wax-removal tail to be summable or vanishing along a pre-registered ε_j ladder.

13 Open Directions

The TLL+LDI framework opens several concrete research directions grouped by theme.

13.1 Pilot verification systems for TLL and LDI

Remark 13.1 (Pilot verification systems for TLL and LDI). Three candidate systems for pilot verification of the TLL and LDI conditions:

1. *Reaction-diffusion (2D):* Smooth nonlinearity, well-understood attractors. TLL and LDI are plausible targets because parabolic smoothing supplies stronger regularity than in the critical Navier–Stokes setting.

2. *Hyperviscous NSE ($(-\Delta)^\beta$, $\beta \geq 5/4$):* A candidate bridge system where inertial-manifold and smoothing estimates may supply the additional tubular branch control needed for TLL. This is a test route, not an automatic TLL guarantee.

3. *Kuramoto–Sivashinsky (1D):* Deterministic chaos without singularities. The attractor is finite-dimensional and smooth, making TLL+LDI verifiable by direct computation.

Remark 13.2 (Numerical sanity check: TLL and LDI on the Lorenz attractor). We verify TLL and LDI numerically on the Lorenz attractor ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$), a 3D chaotic system with fractal attractor dimension ≈ 2.06. The attractor is approximated by 50,000 trajectory points; a test trajectory is initialised with perturbation $(2, 2, 2)$ from an attractor element and integrated for $T = 100$ time units.

Results (the four proof-relevant diagnostics passed; the exponential row is a separate approach-rate diagnostic):

Test	**Result**	**Detail**
TLL (Definition 4.1)	PASS	$C_{\mathrm{TLL}}(95\%) = 0.19$, max $= 1.94$
LDI (Definition 5.1)	PASS	$\int \|u'\|\Omega\, \mathrm{d}t = 6.47 \times 10^4$ (finite)
BV Chain Rule (Theorem 6.1)	PASS	$\mathrm{Var}(v)/(C_{\mathrm{TLL}} \cdot \mathrm{LDI}) = 0.52$
STC (Definition 7.3)	PASS	$\alpha = 1.11 > 0$
Exponential approach fit	DIAGNOSTIC	$\lambda = 0.064 > 0$; not used as an LDI proof

Key observations: (i) the TLL constant at the 95th percentile is $C_{\mathrm{TLL}} \approx 0.19$, but the *maximum* over the test trajectory is 1.94—an order of magnitude larger. Since TLL must hold for *all* t (not 95%), the relevant constant for the BV bound is the maximum. (ii) The LDI/L^1 ratio is 6.86, showing that the log-distance weight inflates the integral by less than one order of magnitude. (iii) The BV chain rule bound is satisfied with margin (ratio $0.52 < 1$). (iv) Sublevel-time control holds with exponent $\alpha \approx 1.1$.

Limitations of the numerical test: The attractor approximation uses 50,000 trajectory points, which systematically overestimates $d(t)$ relative to the true attractor (making TLL constants appear smaller). Moreover, this is a single test at the classical Lorenz parameters

($\sigma = 10$, $\rho = 28$, $\beta = 8/3$) in $\mathbb{R}^3$—a finite-dimensional ODE, not an infinite-dimensional PDE. A systematic parameter scan over ρ (including near-onset values $\rho \approx 24$) would strengthen the evidence. Despite these caveats, the test confirms that the TLL+LDI conditions are not vacuous: they are satisfied in at least one concrete chaotic dynamical system.

Remark 13.3 (Numerical sanity check II: TLL and LDI on the Kuramoto–Sivashinsky attractor)*.* We extend the numerical validation to the Kuramoto–Sivashinsky (KS) equation $u_t + u\,u_x + u_{xx} + u_{xxxx} = 0$ on $[0, L]$ with periodic boundary conditions ($L = 32\pi$), a 1D dissipative PDE with spatiotemporal chaos and finite-dimensional attractor. The pseudo-spectral discretisation uses $N = 128$ Fourier modes; the attractor is approximated by 5,001 points in the $M = 32$-dimensional Fourier coefficient space.

Results (primary run, $N = 128$):

Test	**Result**	**Detail**
TLL (Definition 4.1)	PASS	$C_{\mathrm{TLL}}(95\%) = 9.0$, max $= 58.8$
LDI (Definition 5.1)	PASS	$\int \|u'\|\Omega\,dt = 5576$ (finite), LDI$/L^1 = 1.86$
BV Chain Rule (Theorem 6.1)	PASS	$\mathrm{Var}(v)/(C_{\mathrm{TLL}} \cdot \mathrm{LDI}) = 0.059$
STC (Definition 7.3)	PASS	$\alpha = 0.91 > 0$
Exponential tracking	N/A	$\lambda = -0.074$ (chaotic wandering, no monotone approach)

Grid refinement ($N = 32, 64, 128, 256$; $N = 32$ uses $L = 22$):

N	M	$C_{\mathrm{TLL}}(95\%)$	LDI$/L^1$	BV ratio	STC α	All
32	16	1.81	2.58	0.145	1.85	PASS
64	32	8.75	1.41	0.083	1.20	PASS
128	32	5.10	2.02	0.078	1.11	PASS
256	32	2.66	2.77	0.099	0.79	PASS

Key observations: (i) All four diagnostics pass at every resolution in this test suite, so the TLL/LDI signal is not obviously an artefact of the chosen finite discretisation. (ii) The BV ratio is uniformly small (< 0.15), indicating the chain rule bound is far from tight. (iii) Unlike the Lorenz system ($\mathbb{R}^3$), KS is a *PDE* with infinite-dimensional phase space, truncated to N Fourier modes. The attractor dimension ($\sim$ 8–12 for $L = 32\pi$) is well above the Lorenz value (~ 2.06), making this a substantially harder test. (iv) Exponential tracking fails (expected: the test trajectory wanders *on* the attractor, not toward it). Script: `compute_tll_ldi_ks.py`; results: `ks_tll_ldi_results.json`.

Remark 13.4 (Ginzburg–Landau as TLL/LDI bridge candidate)*.* The complex Ginzburg–Landau equation $\partial_t u = (1 + i\alpha)\Delta u + u - (1 + i\beta)|u|^2 u$ possesses a finite-dimensional inertial manifold for certain parameter ranges. A complete verification of TLL+LDI on this system, followed by a lifting argument to 2D NS (via parameter continuation in the nonlinearity), would provide a concrete test path for the NS-LDI transfer program. It should not be read as a direct route to the full classical 3D NS problem without the additional uniform projection-branch estimates isolated above.

Theorem 13.5 (Conditional squeezing-to-TLL criterion)**.** *Let $u_t + Au + B(u, u) = f$ be a dissipative PDE on a Hilbert space H with compact global attractor $\mathcal{A}$ of finite fractal* *dimension d_F. Suppose the squeezing property holds: there exist $N \in \mathbb{N}$, $\theta \in (0,1)$ and $T > 0$ such that for all u_0, v_0 in the absorbing ball,*

$$\begin{aligned} &\|Q_N(S(T)u_0 - S(T)v_0)\| \le \theta\,\|u_0 - v_0\|, \\ \textit{or} \qquad &\|Q_N(S(T)u_0 - S(T)v_0)\| \le \|P_N(S(T)u_0 - S(T)v_0)\|. \end{aligned}$$

where P_N projects onto the first N eigenmodes of A. Assume in addition that, along the trajectory under consideration, there is a single-valued nearest-point branch $\pi_{\mathcal{A}}$ on a tube $\{x : \operatorname{dist}(x, \mathcal{A}) < r_0\}$ containing the trajectory segment, and that this branch is Lipschitz there with constant L_π. Then the trajectory log-Lipschitz condition (TLL, Definition 4.1) holds on that segment with

$$C_{\mathrm{TLL}} \le \max\{L_\pi, 2\}.$$

If the tubular branch constants are controlled by the squeezing data, one may write this schematically as $C_{\mathrm{TLL}} \le C(N, d_F, \theta, r_0, L_\pi)$.

Proof. The argument proceeds in two steps.

Step 1 (Lipschitz graph). The squeezing property implies that the global attractor $\mathcal{A}$ lies in a Lipschitz graph over the finite-dimensional subspace $P_N H$. This is the Mané projection theorem applied to the squeezing dichotomy: since the high-mode component Q_N contracts relative to the low-mode component P_N, the restriction $P_N|_{\mathcal{A}}$ is injective with Lipschitz inverse [12, 17]. Concretely, there exists $L = L(N, d_F) < \infty$ such that $\|Q_N(u - v)\| \le L\,\|P_N(u - v)\|$ for all $u, v \in \mathcal{A}$.

Step 2 (The missing projection-regularity input). The previous step alone does *not* imply that the nearest-point projection onto $\mathcal{A}$ is Lipschitz: a general Lipschitz graph need not have positive reach or be prox-regular. This is the genuine gap. Under the additional tubular branch assumption, however, for $t, t + h$ in the segment we have

$$\|\pi_{\mathcal{A}}(u(t + h)) - \pi_{\mathcal{A}}(u(t))\| \le L_\pi\,\|u(t + h) - u(t)\|.$$

Since $\Omega(t) \ge 1$, this is exactly TLL with $C_{\mathrm{TLL}} \le L_\pi$ away from $\{d = 0\}$; the convention $C_{\mathrm{TLL}} \ge 2$ handles the set $\{d = 0\}$ as in Remark 4.2. Thus squeezing identifies the finite-dimensional graph route, while prox-regular or otherwise Lipschitz nearest-point branch control is the additional ingredient needed to turn it into TLL. ☐

Remark 13.6 (Hyperviscous Navier–Stokes as conditional TLL/LDI pilot). For the hyperviscous NS equation $\partial_t u + (-\Delta)^\beta u + (u \cdot \nabla)u = -\nabla p + f$ with $\beta \ge 5/4$ on $\mathbb{T}^3$, the known smoothing and inertial-manifold literature make this a strong pilot system for the criterion above. If the relevant inertial-manifold branch provides uniform tubular nearest-point control and independent LDI control (for instance via STC or a lower distance envelope), then one obtains the conditional chain

$$\begin{aligned} \text{IM/prox-regular branch} &\Rightarrow \text{Squeezing + branch control} \\ &\Rightarrow \text{TLL} \Rightarrow \text{TLL + LDI} \Rightarrow \text{BV selection (G')} \\ &\Rightarrow \text{conditional regularity in the companion paper.} \end{aligned}$$

This should be read as a proof-grade test route, not as an unconditional closure of the classical 3D Navier–Stokes problem. For classical NS ($\beta = 1$), the uniform nearest-point branch/TLL input and the corresponding LDI control remain open.

13.2 Mathematical tools for TLL and LDI

Remark 13.7 (Bridge heuristic: squeezing to TLL). The squeezing property (Foias–Temam) states that high modes contract faster than low modes diverge. Combined with determining modes, it can provide estimates of the form

$$d(u(t), \mathcal{A}) \leq Ce^{-\gamma_N t} d(u(0), \mathcal{A}) + C' \sup_{s \leq t} \|P_N u(s) - P_N v(s)\|$$

for a suitable comparison trajectory $v \in \mathcal{A}$. Such estimates are useful input for log-distance bounds, but they do not by themselves prove nearest-point TLL. They must be combined with the tubular projection-branch regularity isolated in Theorem 13.5.

Remark 13.8 (Doubling and Assouad structure). Assouad-type control of the attractor trajectory may provide a route to TLL: if the trajectory $\{u(t) : t \in [0, T]\}$ has sufficiently thin metric structure, then approximate tangent cones can have logarithmic opening bounds of order $C/|\log r|$. This is a plausible sufficient mechanism for TLL, not an equivalence in the present paper. Recent structural results on sets of positive reach [15] provide additional tools for understanding the geometry of attractor neighbourhoods.

Remark 13.9 (LDI as tail-integral equivalence). The LDI condition

$$\int_{\{d>0\}} \|u'(t)\| \log \frac{R}{d(t)} \, \mathrm{d}t < \infty$$

is equivalent to the tail-integral condition

$$\int_0^R \frac{1}{s} \left(\int_{\{0<d\leq s\}} \|u'(t)\| \, \mathrm{d}t \right) \mathrm{d}s < \infty.$$

It relates the weighted integrability of $\log(R/d)$ to the "speed" of the trajectory near the attractor and is more amenable to energy estimates.

Remark 13.10 (Proposition 7.5 clarification). The BV $\to L^\infty$ embedding in Proposition 7.5 requires the additional condition $d(t) > 0$ a.e.: the trajectory must not touch the attractor on a set of positive measure. This is a no-positive-measure-contact assumption, not an automatic Leray–Hopf fact. For strong analytic solutions it may be justified under additional hypotheses, or it may be replaced by Variant A on intervals where the trajectory lies on the attractor.

13.3 Stochastic extensions and advanced tools

Remark 13.11 (Stochastic stabilisation). Adding Feller-type noise $\sigma \, \mathrm{d}W$ to the Navier–Stokes equations may help enforce TLL-type transversality "almost surely": the noise can reduce persistent tangential approach to the attractor, and the Feller property gives continuity of the transition semigroup. This is a heuristic stabilisation route, not a theorem that stochastic forcing automatically supplies a TLL nearest-point branch.

Remark 13.12 (High-mode stability conjecture). A possible route to TLL in the NS context is high-mode stability: if the high Fourier modes $Q_N u = (I - P_N)u$ satisfy uniform decay estimates and the induced nearest-point branch is controlled in the corresponding low-mode skeleton, then TLL may follow from determining-mode estimates plus the tubular branch criterion. High-mode decay alone is not enough, because it controls distance to a low-mode skeleton rather than the regularity of the nearest-point map. Recent sharp dimension estimates for NS-type attractors [14] provide quantitative input for this route.

Remark 13.13 (Variational analysis perspective). The theory of metric regularity and variational analysis (Rockafellar–Wets [19], Dontchev–Rockafellar [8]) provides a useful analytical language for the TLL and LDI conditions: TLL is related to the Aubin property of the attractor projection, and LDI is a calmness condition on the distance function. Tools from variational analysis—including Ekeland's variational principle, the Mordukhovich criterion, and the Attouch–Théra duality—may therefore be relevant in this context.

13.4 Further NS-specific research directions

Remark 13.14 (Further research directions).

1. *Selection regularity via trajectory attractors:* Formulate the minimal metric derivative selection in the Chepyzhov–Vishik trajectory attractor framework, where non-uniqueness is handled by considering entire trajectories rather than initial data.

2. *Quantitative tracking lemmas:* Establish measure estimates $|\{t : d(t) \leq \varepsilon\}| \leq C\varepsilon^\alpha$ via energy/absorbing-ball arguments and "no sticky contact" conditions.

3. *Ekeland variational principle:* Use Ekeland's principle to construct "almost minimisers" with controlled variation, bypassing the need for positive reach of $\mathcal{A}$.

4. *Set-valued BV selection:* Determine which dynamical regularity conditions on the flow force bounded Hausdorff variation of the minimiser set $\{u \in \mathcal{A} : d(u, u(t)) = d_{\mathcal{A}}(u(t))\}$.

5. *Approximate prox-regularity:* Establish a modified reach after removing a meager set (Lebesgue-null): the attractor may fail to have positive reach on a negligible set but retain it generically. Results of Lytchak [15] on the structure of sets of positive reach suggest that such a decomposition may be feasible.

6. *Weak selection functions:* Replace strong Lipschitz projections with selection functions of bounded Hausdorff variation, which exist under weaker regularity assumptions.

7. *Geometric Euler system:* Construct a coherent hierarchy of finite Galerkin approximations $\mathcal{A}_N \to \mathcal{A}$ as a "reach surrogate". One cannot assume positive reach for arbitrary projected attractor point clouds; the required input is instead a verified or regularised skeleton branch with controlled reach/prox-regularity and Hausdorff or Mosco convergence.

13.5 Cross-paper connections

Remark 13.15 (DFC from turbulence as topological funnel heuristic for TLL). The Downhill Flux Condition (DFC) from the turbulence companion paper [13] expresses that energy flows "downhill" through the inertial range. Translated to the attractor geometry, this suggests a topological funnel picture: DFC may constrain the approach direction and thereby support a log-Lipschitz rate. This is a physical heuristic for TLL, not a replacement for the tubular projection-branch estimate required in Theorem 13.5.

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